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Principles of Environmental Sciences

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BOOK REVIEWS

Principles of Environmental Sciences, edited by Jan J. Boersema and Lucas Reijnders, Dordrecht/Boston, Springer 2009, 542 pp., €99.95, ISBN 978-1-4020-9157-5 (hardback)

Principles of Environmental Sciences is an academic textbook that fits the bookshelves of all students in environmental sciences. The aim of the book is (1) to describe environmental problems in their historical context, (2) to delineate how complex environmental problems can be analysed and tackled by using various (inter)disciplinary concepts, methods and tools and, (3) to illustrate how solutions work out in their social context. The learning aims are formalized as follows by the editors: "*We expect students to learn some specific skills, e.g. the essentials of building a model or applying LCA, but our main objective is to improve their ability to analyse and conceptualise environmental problems in context, to make students aware of the value and scope of different methods and to teach them the results and insights of previous work in this field*".

The book consists of 28 chapters, and is subdivided into three parts: Part I, *Stating the Problem* (6 chapters); Part II, *Principles and Methods* (12 chapters), and Part III, *Context and Perspectives* (10 chapters). Part I includes an introduction to environmental sciences, and an overview of the historical context. Part II is the core of the book and focuses on guiding principles and disciplinary and multidisciplinary methods. The text includes practical examples. Special emphasis is given to the concept of integration, modelling, and integrated assessment. Part III illustrates how solutions work in a specific societal context. It includes case studies on different spatial scales. These case studies illustrate how (technical/practical) solutions of a more or less similar problem may provoke quite different reactions in different societies. Finally, the book offers perspectives on economic growth and on major societal sectors and the most likely course they will take in the future.

The chapters are written by 37 contributors from Europe and North America. Natural and social sciences are evenly distributed among the contributors. Their backgrounds range from theoretical physics, to ecology, psychology, sociology and many other disciplines.

The book is primarily intended for students in environmental science: for undergraduate and graduate students of multi- and interdisciplinary courses in environmental studies/sciences and courses focusing on methodology, and for graduate students specialising in environmental topics of their discipline.

The editors aimed at a book that distinguishes itself from other textbooks for two main reasons. The first is that the book pays considerable attention to environmental history and to the variations in both perception and the implementation of solutions to environmental problems. The second is the focus on principles, concepts and methodology.

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Indeed, the book includes a number of valuable chapters on the historical and cultural context of environmental issues. For instance, Part I of the book describes environmental changes over the last 3 million years, and chapters on environmental history and recent developments and trends. Part II includes a chapter on social science and environmental behaviour, and Part III a number of interesting examples of environmental policies in their cultural and historical context.

Why I think this book is valuable for our students, is especially because of the methodological focus. Part II of the book includes a comprehensive and well-written overview of methods and tools applied in environmental sciences, such as cost-benefit analysis, material flow analysis, life-cycle assessment, environmental modeling and Integrated Assessment. Moreover, it describes general and specific principles, starting with the principle of Sustainable Development stating that "*We humans must move towards the adoption of a lifestyle that can continue indefinitely*" (Chapter 7, by Craig Dilworth).

As a result of the focus on methodology, the book does not introduce the causes and effects of environmental problems in depth. There are no detailed explanations of problems such as global warming, ozone depletion, acidification, eutrophication or biodiversity loss. Instead, these problems, and their solutions, are addressed through illustrative examples and specific discussions. The editors suggest that the students should have some basic knowledge of environmental science, or some years of disciplinary training. The book can nevertheless be useful for students with little background in environmental science. The chapters are well written, and generally start from the basics, assuming little basic knowledge of the issue at stake.

Many environmental problems are complex and of an international nature. Therefore, the environmental scientists of the future should be able to cooperate with scientists from other disciplines in international projects. They should also be able to understand the political and societal context of the problems. The book *Principles of Environmental Sciences* provides a basis to train such scientists.

The strength of the book is that it considers environmental problems as resulting from observed phenomena in our natural environment on one hand, and the societal awareness of these phenomena on the other. From this starting point, the principles and methodologies are introduced to the reader. The editors succeeded in their aim to "*avoid the 'Scylla of positivism' (as natural scientists, we know what the problems are) and the 'Charybdis of constructivism' (problems only exist if we see them as problems)*". In Greek mythology, the two sea monsters Scylla and Charybdis were on opposite sides of a narrow strait to be passed by ships, presumably similar to the narrow Strait of Messina. Their position made the Strait difficult to pass for sailors. Some, however, succeeded with the help of others. The editors of *Principles of Environmental Sciences* invited many others to contribute to the book. Their excellent selection of authors resulted in chapters giving a wide range of disciplinary viewpoints, and providing a solid basis for teaching the basics of environmental sciences.

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